1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In tumors with high stromal-to-tumor ratio (TSR), TP53 deficiency creates reliance on PROS1-mediated signaling to sustain proliferative clones via TAM receptor pathways. PROS1, produced by tumor and/or stromal cells, engages TAM receptors (TYRO3, AXL, Mer) on cancer and/or stromal cells, amplifying downstream proliferative signaling (e.g., AKT/ERK) and supporting clonal persistence after local therapy. This axis is strengthened in TSR-high environments due to enhanced stromal support and paracrine PROS1 availability. Consequently, tumors that are TSR-high, TP53-mutant, and PROS1-high will exhibit accelerated clonal expansion, higher local recurrence after treatment, and shorter progression-free interval (PFI). MKI67 (Ki-67) expression, while correlated with proliferation, acts as a secondary modifier that modulates magnitude rather than being essential to the core axis. Positive surgical margins and the index lesion contribute to microenvironmental context, increasing the likelihood that PROS1-mediated signaling promotes early progression but are not required for the axis to operate.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TP53, PROS1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53-altered tumors recur and progress rapidly while others do not — specifically, what tumor-microenvironment features enable TP53-driven proliferation to translate into early recurrence/progression? 2) Tier 1 paths used: (a) Proliferation ↔ TP53 (mechanistic backbone linking TP53 dysfunction to deregulated cell division) and (b) PROS1 ↔ Ki-67 (connects PROS1 expression to tumor proliferative phenotype).

PROS1: PROS1 ↔ Ki-67 (connects PROS1 expression to tumor proliferative phenotype).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MKI67	Supporting modifier from knowledge graph

Secondary	Positive surgical margins	Supporting modifier from knowledge graph
Secondary	Index lesion	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: TSR-high, TP53-mutant tumors will show higher PROS1 signaling activity and TAM receptor engagement than TSR-low counterparts;

Prediction 2: In vitro, PROS1 knockdown or TAM axis blockade will attenuate proliferation of TP53-deficient cells in stromal-rich settings;

Prediction 3: In vivo, disrupting PROS1-TAM signaling will reduce post-therapy clonal regrowth in TSR-high, TP53-mutant models;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MKI67: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Positive surgical margins: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.